

# From annotation to summary video

## Group 9

TVSUM videos 33 — 36

We built a system that watches a video, picks the moments that matter, and stitches them into a short clip.

## Four very different videos

### Video 33

#### Graduation flash mob

speech + dance, 7 min, lots of dialogue

### Video 34

#### ICC street flash mob

fast group dance, 2.5 min, no speech

### Video 35

#### Train classical-music flash mob

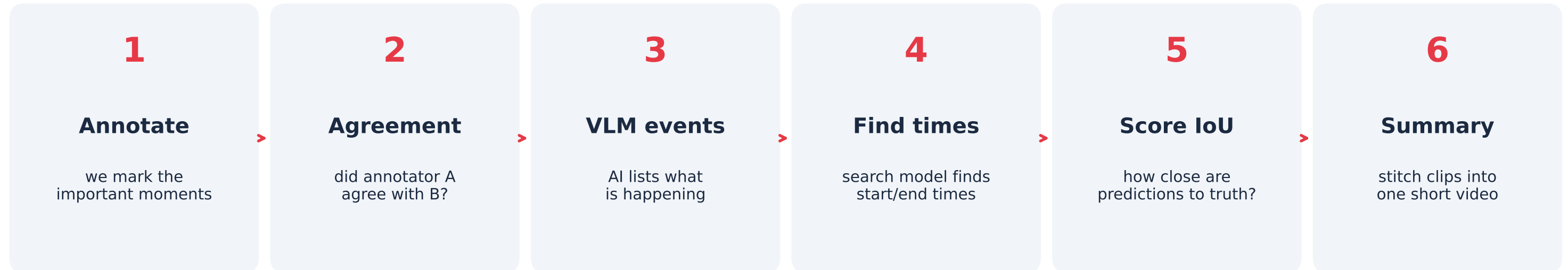
musicians revealing themselves, 2.5 min

### Video 36

#### Beekeeper interview

instructional voice-over, 4 min, no climax

# Six steps, one notebook



*Built as a single notebook + a small Python package*

## Two annotators rarely fully agree

| Video | Same moments?<br>(timeline $\kappa$ ) | Same boundaries?<br>(mean IoU) | Same importance?<br>(saliency $\kappa$ ) |
|-------|---------------------------------------|--------------------------------|------------------------------------------|
| 33    | 0.15                                  | 0.65                           | 0.60                                     |
| 34    | 0.48                                  | 0.70                           | 0.36                                     |
| 35    | 0.38                                  | 0.55                           | 0.23                                     |
| 36    | 0.25                                  | 0.72                           | −0.24                                    |

- Boundaries agree well (0.55–0.72) — when we both pick a moment, we draw similar windows.
- Counts of events disagree more — one of us is a 'splitter', the other a 'joiner'.
- Importance is the most subjective — beekeeper video is below chance.

## Three ways of asking the same question

### Just give me events

Most reliable.  
4–5 events parsed  
per video.

✓ **best overall**

### Events with timestamps

Almost always fails.  
The model has no  
idea when things happen.

✗ **avoid**

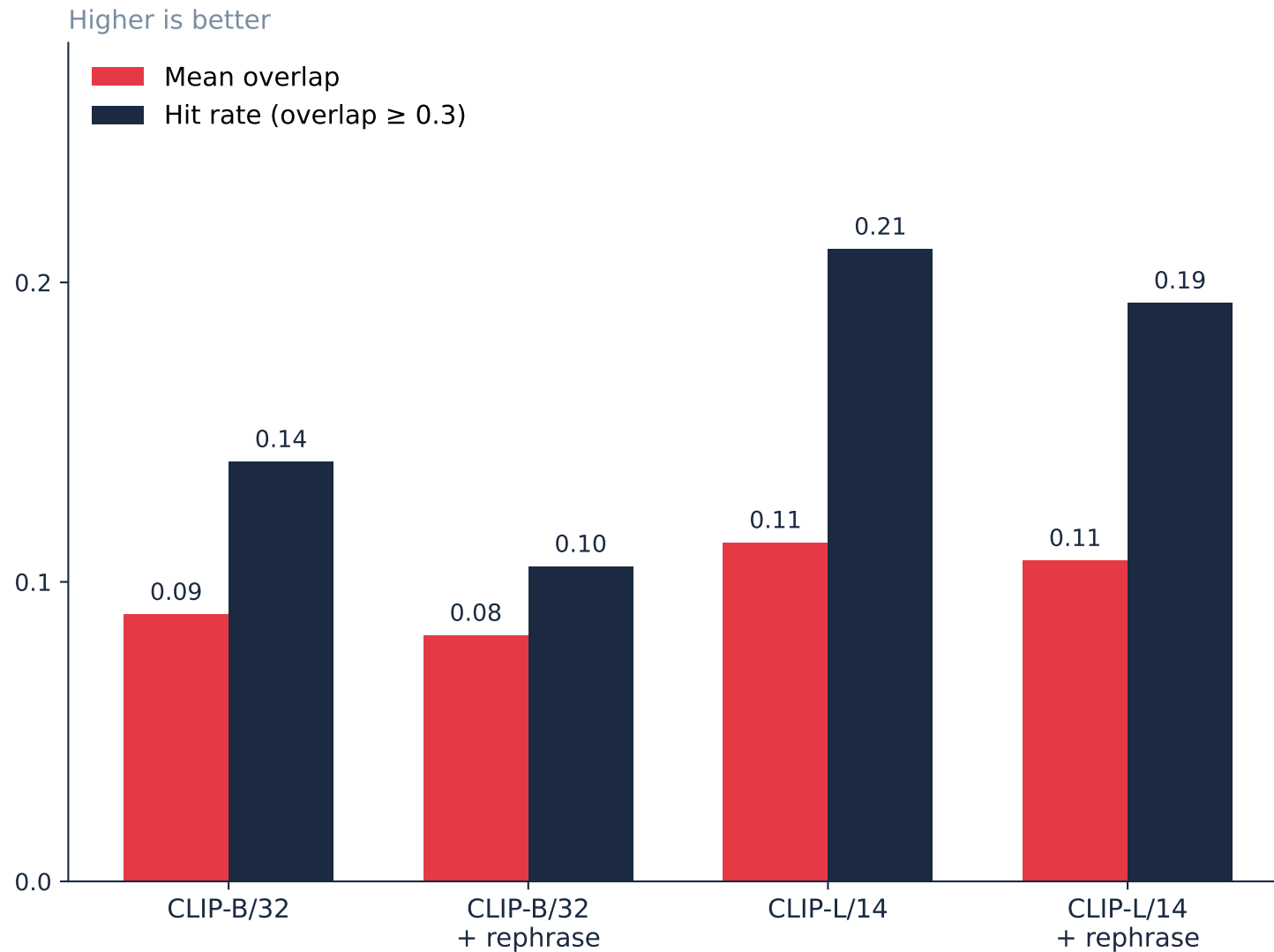
### Caption frames, then merge

Hit-or-miss.  
Works on the beekeeper  
video (covers 42 %).

≈ **situational**

**Big failure: anything that depends on speech is invisible to the model.**

# A bigger model is the cheapest improvement



- Bigger CLIP wins by 27 % overlap with no other change.
- Rephrasing the query 3× actually hurt — averaging blurs the strongest signal.
- By event type: performance 0.17, beekeeping 0.07, speech 0.00

# Walk-through: the train orchestra

## What we annotated (13 events)

- 00:17 — Sign reads 'Classical special train'
- 00:39 — A flutist begins playing
- 00:48 — A man on the oboe joins
- 01:00 — A man with a violin stands up
- 01:24 — Full orchestra plays, crowd reacts
- 01:41 — Conductor stands and conducts
- 02:05 — Train pulls in, performance ends

## What our system did

VLM listed 4 salient events:

1. A person plays a flute on a train
2. A group plays violins together
3. Passengers watch and record
4. Train arrives at the station

CLIP-L/14 located these 4 events:

**Mean overlap: 0.333 (best of all 4 videos)**

**Hit rate at overlap  $\geq 0.3$ : 75 %**

Generated summary: 28 s, 8 clips, in order

*Misses: conductor, closing text overlay*

# Three things we learned

## 1. Subjectivity matters more than model choice.

Two humans annotating the same video already disagree about what is essential. Any AI will reflect \*one\* of those views.

## 2. Vision-only models are speech-blind.

Performances and dance: works.  
Speeches, jokes, narration: zero overlap. Need ASR to fix.

## 3. Bigger model = cheapest win.

Swapping CLIP-B for CLIP-L gave +27 % overlap with no other change. Compute is cheaper than engineering effort.



## NEXT STEPS

# What we'd do with another week

### Add speech recognition (Whisper)

Biggest single win: dialogue events currently score zero overlap; ASR makes them retrievable.

### Re-rank clips by importance

Use saliency ratings as weights so the summary favours essential moments over visually flashy ones.

### Bring back CG-DETR

The model the assignment originally suggested. Its pretrained checkpoint is currently offline, but it would beat zero-shot CLIP by ~25 IoU points.

QUESTIONS?

**Thank you**

## **Group 9 · TVSUM 33-36**

Annotation → VLM → Retrieval → Summary

*Headline: subjectivity is the dominant signal,  
speech is the bottleneck, bigger models help.*